IE6600 COMPUTATION AND VISUALIZATION · HACKATHON · TEAM 4

Clean Energy Got Cheap. Did We Switch?

How falling solar and wind costs compare with how fast the US, and the world, actually moved away from fossil power

Taniya

Nehal

Tyler Brown

–84%

cost of new US solar since 2009

19%

of US electricity from wind and solar (2025)

10th of 19

US rank in the G20 for cutting fossil power

Prof. Dehghani · Fall 2026

? THE QUESTION AND OUR BIG IDEA

Clean power is now the cheapest new power. Has the US switched to it?

OUR BIG IDEA

Solar and wind now cost 84% and 55% less than in 2009 and are cheaper than coal and gas, yet the US gets only 19% of its electricity from them, compared with about 45% in Portugal and the Netherlands.

Audience: state energy officials deciding which power plants get built next.

Action: plan new capacity around solar and wind, and measure progress against the leaders, not the world average.

Q1 COST

How much have solar and wind costs fallen compared with coal and gas?

Data: Lazard LCOE+ (US) · IRENA (world)

Q2 MIX

How fast has the electricity mix changed in the US and worldwide?

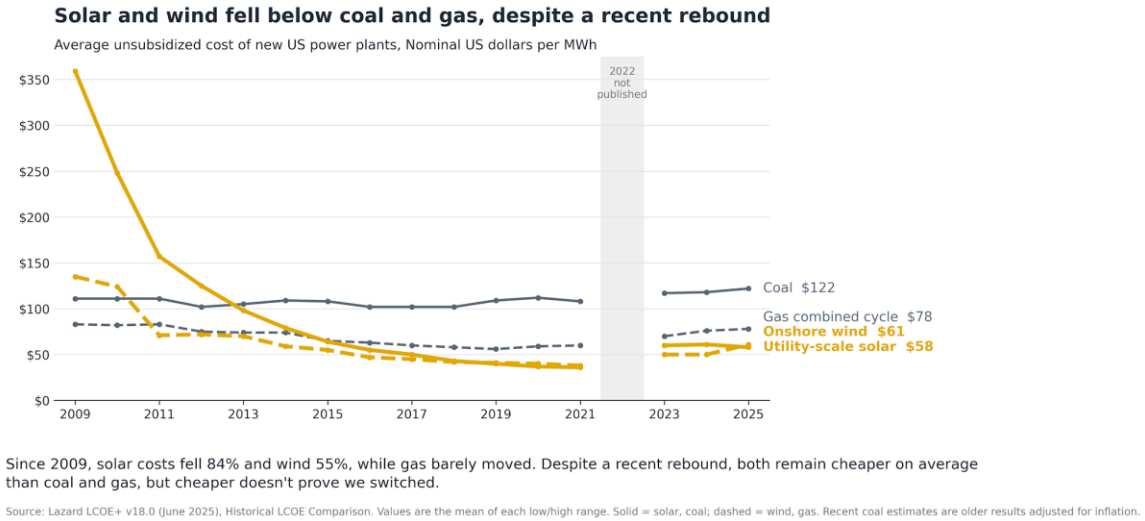
Data: Our World in Data (Ember; Energy Institute)

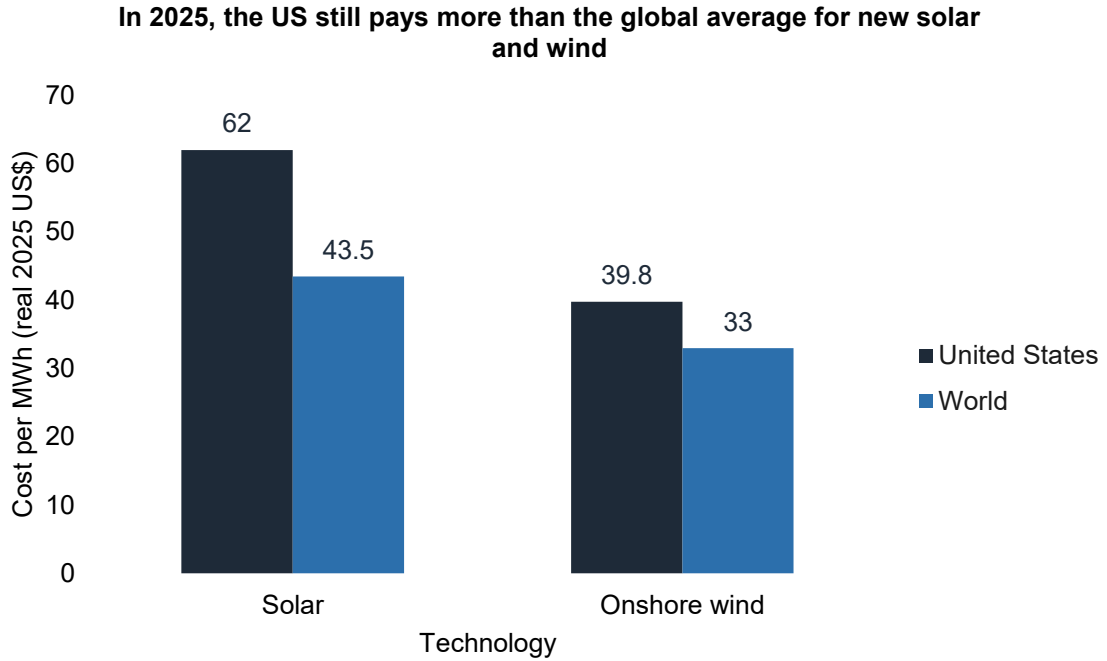
Q3 PACE

Which countries are switching away from fossil fuels fastest, and is the US among them?

Data: Our World in Data energy dataset

Solar and wind are now the cheapest new US power, but the US still pays more than the world average





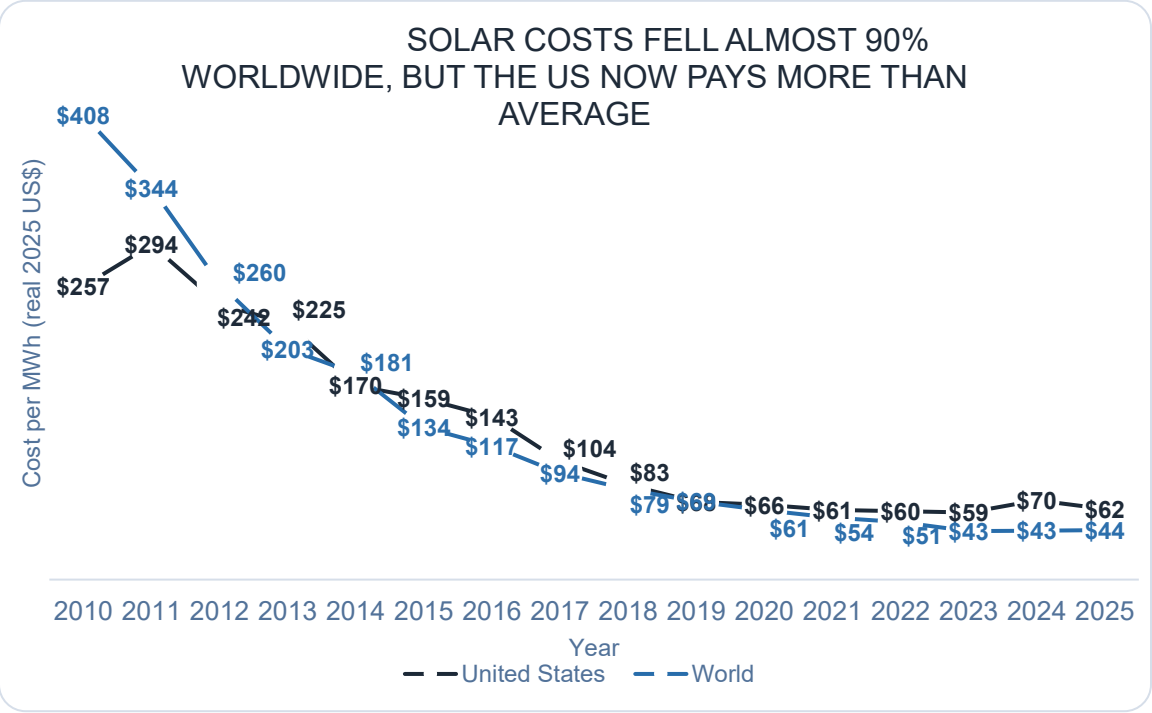
Since 2009 the average cost of new US solar fell 84% and onshore wind 55% (nominal $). Despite a rebound after 2021, both were cheaper than gas ($78) and coal ($122) per MWh in 2025.

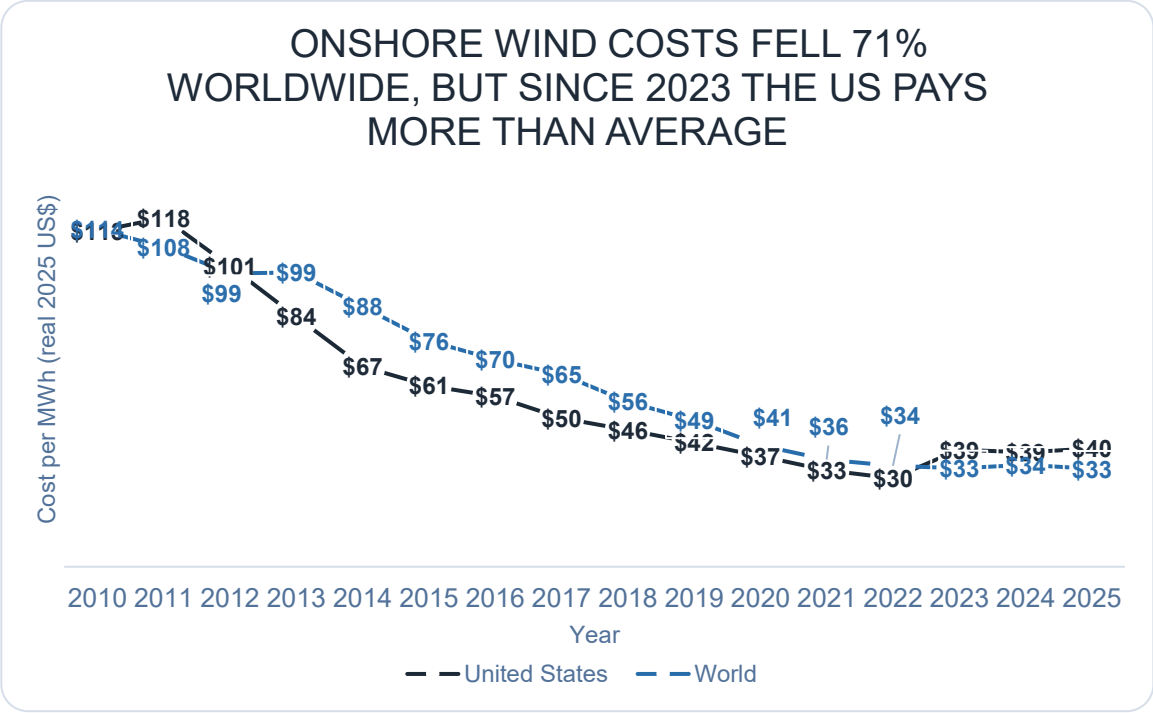
Even after the global cost collapse, new US solar costs $62 per MWh against a $44 global average (about 40% more), and onshore wind $40 against $33 (about 20% more).

Python · workbook tab: Q1 Python · Source: Lazard LCOE+ v18.0 (June 2025)

Excel · workbook tab: Q1 IRENA 2025 Bar · Source: IRENA (2026), real 2025 US$

The cost collapse is global: since 2010, solar fell about 89% and onshore wind about 71% worldwide





The global average cost of new solar fell from $408 to $44 per MWh (−89%); the US fell from $257 to $62 (−76%) and has cost more than the world average since 2020.

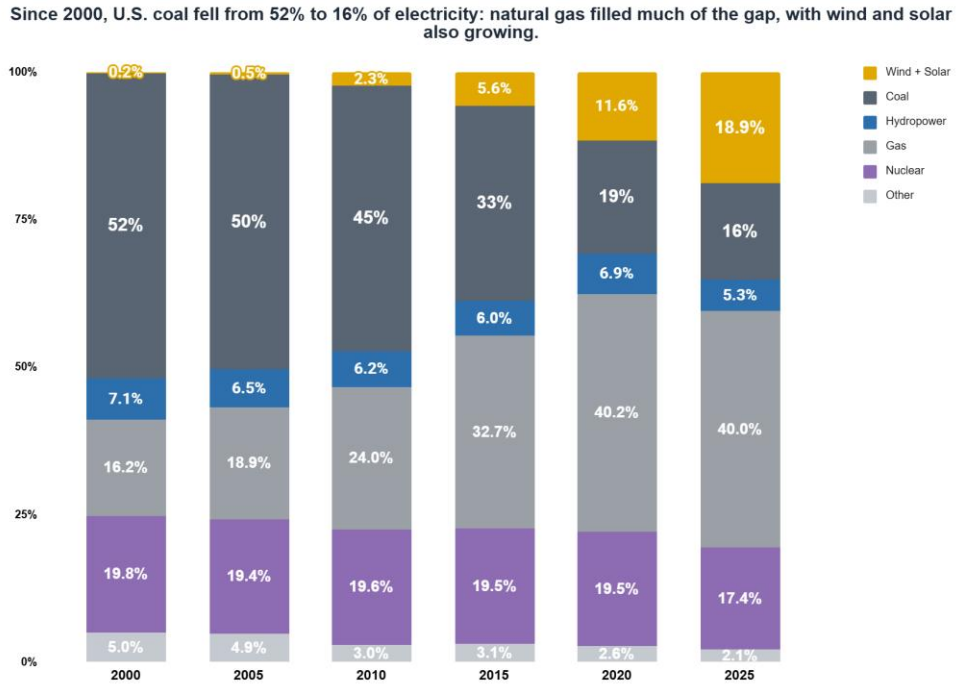
Excel · workbook tab: Q1 IRENA Solar Pivot · Source: IRENA (2026), real 2025 US$

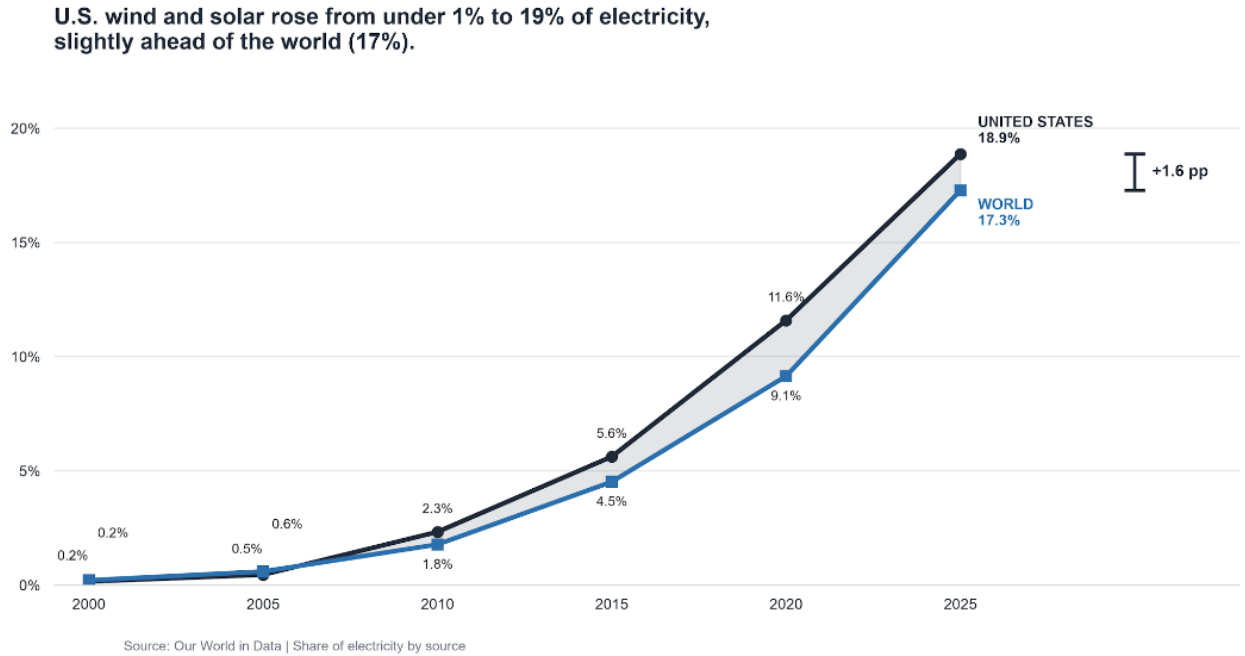
Onshore wind fell from $114 to $33 per MWh worldwide (−71%). US wind was cheaper than average for most of the decade, but rose about a third after 2022 to end about 20% above it.

Excel · workbook tab: Q1 IRENA Wind Pivot · Source: IRENA (2026), real 2025 US$

If clean power is now the cheapest, the US should be switching fast. Is it? →

The US did switch, but mostly from coal to gas; wind and solar reached 19%, just ahead of the world





Coal fell from 52% to 16% of US electricity between 2000 and 2025. Gas rose from 16% to 40% and wind + solar from under 1% to 19%: much of the move away from coal went to another fossil fuel.

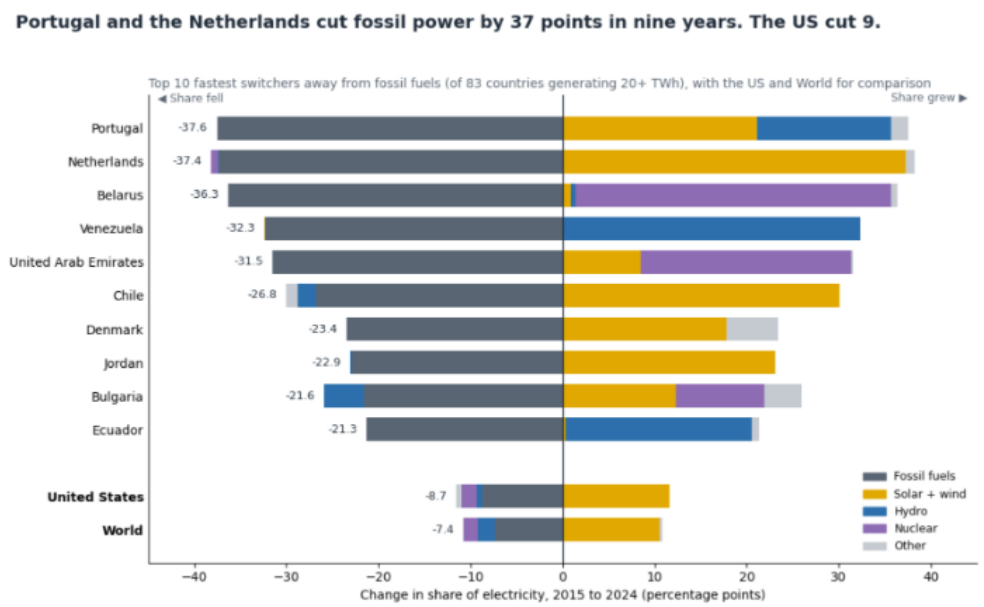
US wind + solar (18.9%) is only 1.6 points above the world (17.3%). The world figure is weighted by generation; the median of 88 countries is 13.1%, so the US is ahead of about 60% of them.

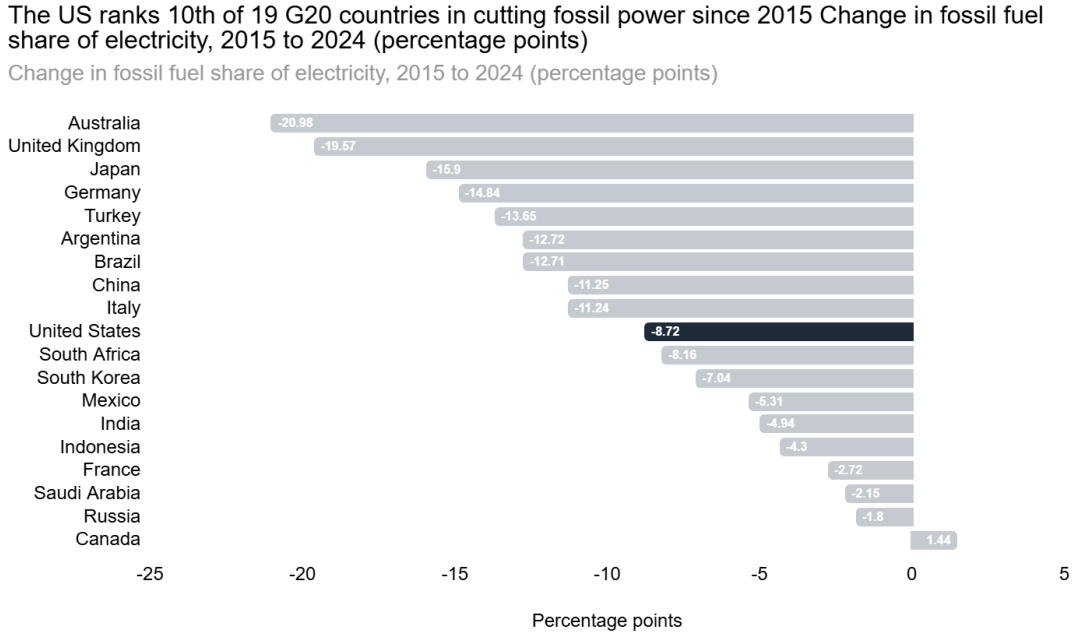
Spreadsheet chart · workbook tab: Q2 Electricity Mix · Source: Our World in Data (Ember; Energy Institute)

Python · workbook tab: Q2 Python Chart · Source: Our World in Data (Ember; Energy Institute)

Ahead of the average, but how does the US compare with the fastest switchers? →

The leaders cut fossil power about four times faster than the US, and by more than one route





Of 83 countries generating 20+ TWh, Portugal (−37.6 pts) and the Netherlands (−37.4) cut fossil share most in 2015–24: through solar and wind, nuclear (Belarus, UAE) or hydro (Ecuador). The US cut 8.7, the world 7.4.

Python · workbook tab: Python Graph · Source: Our World in Data (Ember; Energy Institute)

The US ranks 10th of 19 G20 countries (66.9% → 58.1% fossil), slightly faster than the world but behind Australia (−21.0), the UK (−19.6), Japan and Germany. Canada's share rose a little from a hydro-heavy base.

Excel · workbook tab: Q3 G20 · Source: Our World in Data (Ember; Energy Institute)

Cheap clean power has not yet meant a fast switch in the US

Q1

−84%

cost of new US solar since 2009; solar and wind are now cheaper than coal and gas, though about 20–40% above the global average

Q2

19%

of US electricity came from wind and solar in 2025; most of coal's decline went to gas

Q3

~1 pt / yr

US cut in fossil share (2015–24), against ~4 pts a year for the leaders: 10th of 19 in the G20

WHAT THIS MEANS FOR STATE ENERGY OFFICIALS

1. Cost is no longer the barrier: treat solar and wind as the default least-cost option when planning new plants.
2. Benchmark progress against the leaders (about 4 points a year), not the world average (under 1 point a year).
3. Find out why US projects cost 20–40% more than the global average (permitting, grid connection, financing): the next question our data cannot answer.

What our data can and cannot tell us

Question and data	Assumptions	Limitations
Q1 Cost Lazard LCOE+ v18.0: US, 2009–2025, nominal $. IRENA (2026): world and 30+ countries, 2010–2025, real 2025 $.	Levelized cost of new plants (unsubsidized, mean of Lazard's low–high range) is a fair proxy for what new power costs. IRENA's global figure is a generation-weighted average.	Lazard is nominal: after inflation gas fell ~37% and coal ~27%, not flat. Lazard skipped 2022. IRENA covers renewables only, so the two sources are never compared directly. Plant costs exclude grid, storage and integration costs.
Q2 Mix Our World in Data (Ember; Energy Institute): share of electricity generation by source, 2000–2025.	'Other' = oil + bioenergy + other renewables; 'Wind + solar' = sum of the two shares. Mix shown every five years. Median taken across the 88 countries with 2025 data.	Electricity only, not total energy use. Shares, not absolute generation. 2025 values are partly preliminary. The world figure is weighted by generation, so large producers dominate it.
Q3 Pace Our World in Data energy dataset (2026 release): electricity shares by source, 2015 and 2024.	Pace = 2024 minus 2015 fossil share (percentage points). Only countries generating 20+ TWh in 2024 (83). Blank hydro (Saudi Arabia, UAE) and hydro and wind (Singapore) set to 0: each country's low-carbon total minus its listed sources is exactly 0.00 every year.	Two endpoints hide year-to-year swings (hydro depends on rainfall). Ukraine excluded: its data ends in 2022, so its 2015–2024 change can't be calculated. Venezuela's fall reflects economic crisis. Shares, not emissions. Shows who switched, not why.

All three questions show patterns over time, not causes: falling costs and faster switching move together, but our data cannot prove one caused the other. Every dataset is in the workbook with its link (see the ReadME tab).

Thank you. Questions?

Scan the code to evaluate our presentation

TEAM 4

Clean Energy Got Cheap. Did We Switch?

DID YOU KNOW?

Wind + solar passed nuclear in 2025: 18.9% vs 17.4% of US electricity, the largest low-carbon source for the first time.

Cheap isn't enough: of 53 US wind, solar and geothermal projects delayed or blocked in 2008–2021, 49% were cancelled for good, losing about 4,600 MW.

The grid is the queue: 2,061 GW of new generation and storage was waiting to connect to the US grid at the end of 2025; recent projects waited over 4 years.

Sources: our Q2 data (Our World in Data); Susskind et al., Energy Policy 165 (2022); Berkeley Lab, Queued Up (2026 edition).



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